

Quiz 5

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S S M T W T F

Answer 1

Problem: DHCP uses broadcast message (DHCP Discover). This message cannot pass through routers. Since Zahin's DHCP server is on another network Dipu's device cannot reach it to obtain IPv4 configuration.

Solution: Configure a DHCP Relay Agent (IP Helper Address) on Dipu's router so that DHCP requests are forwarded to Zahin's DHCP server.

Answer 2

When I connect to the coffee shop's Wi-Fi my device automatically sends a DHCP request. The coffee shop's DHCP server then automatically provides the necessary IP configuration information such as IP address, subnet mask, default gateway and DHCP server. Therefore, no one needs to manually give me an IP address to access the Internet.

Answer 4

The web server uses a private IP address (192.168.1.2). It cannot be directly accessed directly from the internet. To make communication possible R2 uses NAT with port forwarding.

When dip: Dipu sends a request to the public IP address 201.113.13.221 on port 8080, router R2 translates the destination address and port to the private web server address 192.168.1.2:80.

Translation performed by R2: 201.113.13.221:8080 → 192.168.1.2:80

Thus, even though the web server does not have a fixed public IP address, external users can still access it through the public IP and forwarded port configured on R2.